

ITAI 5153 Project 1: Planning and Search - Experiment Results

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Setup

```
In [1]: import algorithms
        from solver import Solver
        from time import time
```

```
In [2]: # Initial example solver
        slv = Solver(algorithms.bfs)
        slv.cube.print_cube()
        print("Solved:", slv.is_solved())
```

Current cube state: WWWWGRRRBBBOOGRRRBBBOYYYYY

```
      W W
      W W
G G  R R  B B  O O
G G  R R  B B  O O
      Y Y
      Y Y
```

Solved: True

```
In [3]: # Example randomization
        slv.scramble(depth=5, rand_seed=42)
```

Current cube state: WBWGBBOWORYRGRGYBRYOWOYG

```
      W B
      W G
B B  O W  O R  Y R
G R  G Y  B R  Y O
      W O
      Y G
```

```
Out[3]: ((0, 3, 0, 1, 3, 3, 4, 0, 4, 2, 5, 2, 1, 2, 1, 5, 3, 2, 5, 4, 0, 4, 5, 1),
        ['u', 'F', 'F', 'u', 'R'])
```

```
In [4]: slv.is_solved()
```

Out[4]: False

Experiment

```
In [5]: def test_solver(scramble_depth: int):
        print("=" * 50)
        print()
        print("Testing with scramble depth =", scramble_depth)

        bfs_slv = Solver(algorithms.bfs)
        scramble_state, scramble_moves = bfs_slv.scramble(depth=scramble_depth, rand_seed=42)
        print("Scramble moves:", scramble_moves)

        print("-" * 50)
```

```

print("Solving with BFS...")
start_time = time()
bfs_solution, bfs_nodes = bfs_slv.solve()
bfs_time = time() - start_time
print("BFS solution:", bfs_solution)
print("Cost:", len(bfs_solution))
print("Nodes expanded:", bfs_nodes)
print("Time:", bfs_time)

print("-" * 50)
print("Solving with DFS (depth limit 15)...")
dfs_slv = Solver(algorithms.dfs, initial_state=scramble_state)
start_time = time()
dfs_solution, dfs_nodes = dfs_slv.solve(max_depth=15)
dfs_time = time() - start_time
print("DFS solution:", dfs_solution)
print("Cost:", len(dfs_solution))
print("Nodes expanded:", dfs_nodes)
print("Time:", dfs_time)

print("-" * 50)
print("Solving with A*...")
a_star_slv = Solver(algorithms.a_star, initial_state=scramble_state)
start_time = time()
a_star_solution, a_star_nodes = a_star_slv.solve()
a_star_time = time() - start_time
print("A* solution:", a_star_solution)
print("Cost:", len(a_star_solution))
print("Nodes expanded:", a_star_nodes)
print("Time:", a_star_time)

print()

```

In [6]: depths = [0, 1, 3, 5, 10, 15, 25, 100]

```

for depth in depths:
    test_solver(depth)

```

=====

Testing with scramble depth = 0
Current cube state: WWWGGRRBBBOGGRRBBBOYYYY

W W
W W
G G R R B B O O
G G R R B B O O
Y Y
Y Y

Scramble moves: []

Solving with BFS...

BFS solution: []

Cost: 0

Nodes expanded: 0

Time: 1.9073486328125e-06

Solving with DFS (depth limit 15)...

DFS solution: []

Cost: 0

Nodes expanded: 0

Time: 1.430511474609375e-06

Solving with A*...

A* solution: []

Cost: 0

Nodes expanded: 0

Time: 7.152557373046875e-07

=====

Testing with scramble depth = 1

Current cube state: WWWWOOGRRBBGGRRBBBOYYYY

W W
W W
O O G G R R B B
G G R R B B O O
Y Y
Y Y

Scramble moves: ['u']

Solving with BFS...

BFS solution: ['U']

Cost: 1

Nodes expanded: 1

Time: 2.9802322387695312e-05

Solving with DFS (depth limit 15)...

DFS solution: ['F', 'F', 'F', 'R', 'F', 'F', 'F', 'U', 'F', 'F', 'u', 'r', 'F', 'r', 'U', 'F']

Cost: 15

Nodes expanded: 112161

Time: 0.17924761772155762

Solving with A*...

A* solution: ['U']

Cost: 1

Nodes expanded: 2

Time: 4.982948303222656e-05

=====

Testing with scramble depth = 3

Current cube state: WWYYOBRRGRBBGRGGOBOWWYY

```
W W
Y Y
O B R R G R B B
G R G G O B O O
W W
Y Y
```

Scramble moves: ['u', 'F', 'F']

Solving with BFS...

BFS solution: ['F', 'F', 'U']

Cost: 3

Nodes expanded: 8

Time: 6.031990051269531e-05

Solving with DFS (depth limit 15)...

DFS solution: ['F', 'F', 'F', 'R', 'F', 'F', 'F', 'R', 'f', 'r', 'U', 'R', 'u', 'F', 'r']

Cost: 15

Nodes expanded: 68908

Time: 0.1100008487701416

Solving with A*...

A* solution: ['F', 'F', 'U']

Cost: 3

Nodes expanded: 7

Time: 0.00011467933654785156

=====

Testing with scramble depth = 5

Current cube state: WBWGBBOWORYRGRGYBRYOWOYG

```
W B
W G
B B O W O R Y R
G R G Y B R Y O
W O
Y G
```

Scramble moves: ['u', 'F', 'F', 'u', 'R']

Solving with BFS...

BFS solution: ['r', 'U', 'F', 'F', 'U']

Cost: 5

Nodes expanded: 551

Time: 0.02061152458190918

Solving with DFS (depth limit 15)...

DFS solution: ['F', 'F', 'F', 'R', 'R', 'R', 'F', 'U', 'f', 'U', 'f', 'u', 'R', 'f', 'U']

Cost: 15

Nodes expanded: 14165569

Time: 22.890400648117065

Solving with A*...

A* solution: ['r', 'U', 'F', 'F', 'U']

Cost: 5

Nodes expanded: 73

Time: 0.0012271404266357422

=====

Testing with scramble depth = 10

Current cube state: GRORRBYWGBWYGBYWOWOROYG

```

      G R
      O R
R B  Y W  G B  W Y
G B  Y B  W O  W O
      R O
      Y G

```

Scramble moves: ['u', 'F', 'F', 'u', 'R', 'R', 'R', 'u', 'F', 'u']

Solving with BFS...

BFS solution: ['F', 'F', 'r', 'F', 'u', 'f']

Cost: 6

Nodes expanded: 737

Time: 0.006231784820556641

Solving with DFS (depth limit 15)...

DFS solution: ['F', 'F', 'F', 'R', 'F', 'F', 'R', 'U', 'R', 'F', 'F', 'F', 'R', 'F']

Cost: 14

Nodes expanded: 583142

Time: 0.8774867057800293

Solving with A*...

A* solution: ['f', 'f', 'r', 'F', 'u', 'f']

Cost: 6

Nodes expanded: 318

Time: 0.006689548492431641

=====

Testing with scramble depth = 15

Current cube state: ROOGWGWYRWBBGBYOYGWORBYR

```

      R O
      O G
W G  W Y  R W  B B
G B  Y O  Y G  W O
      R B
      Y R

```

Scramble moves: ['u', 'F', 'F', 'u', 'R', 'R', 'R', 'u', 'F', 'u', 'u', 'r', 'F', 'r', 'f']

Solving with BFS...

BFS solution: ['F', 'R', 'f', 'R', 'U', 'F', 'F', 'r', 'F', 'u', 'f']

Cost: 11

Nodes expanded: 628695

Time: 10.614166498184204

Solving with DFS (depth limit 15)...

DFS solution: ['F', 'F', 'F', 'U', 'R', 'F', 'R', 'u', 'f', 'U', 'U', 'F', 'u', 'r', 'F']

Cost: 15

Nodes expanded: 70004192

Time: 111.82055854797363

Solving with A*...

A* solution: ['u', 'F', 'r', 'u', 'r', 'F', 'u', 'F', 'u', 'r', 'U']

Cost: 11

Nodes expanded: 260818

Time: 10.227304697036743

=====

Testing with scramble depth = 25

Current cube state: BYBBYWROWRGRGOYGORGOWBYW

```

      B Y
      B B

```

Y W R O W R G R
G O Y G O R G O
B W
Y W

Scramble moves: ['u', 'F', 'F', 'u', 'R', 'R', 'R', 'u', 'F', 'u', 'u', 'r', 'F', 'r', 'f', 'R',
'R', 'F', 'r', 'u', 'u', 'u', 'r', 'f', 'R']

Solving with BFS...

BFS solution: ['F', 'F', 'r', 'U', 'R', 'R', 'f', 'r', 'F', 'r', 'f']

Cost: 11

Nodes expanded: 563179

Time: 9.320245027542114

Solving with DFS (depth limit 15)...

DFS solution: ['F', 'F', 'F', 'R', 'R', 'R', 'f', 'U', 'r', 'U', 'r', 'f', 'R', 'R', 'f']

Cost: 15

Nodes expanded: 15032703

Time: 24.51888680458069

Solving with A*...

A* solution: ['U', 'R', 'u', 'F', 'r', 'f', 'U', 'f', 'r', 'F', 'F']

Cost: 11

Nodes expanded: 261832

Time: 10.882238149642944

=====

Testing with scramble depth = 100

Current cube state: OWRRYGYWGRBBGBWRBG000YYW

O W
R R
Y G Y W G R B B
G B W R B G O O
O Y
Y W

Scramble moves: ['u', 'F', 'F', 'u', 'R', 'R', 'R', 'u', 'F', 'u', 'u', 'r', 'F', 'r', 'f', 'R',
'R', 'F', 'r', 'u', 'u', 'u', 'r', 'f', 'R', 'f', 'r', 'U', 'F', 'R', 'u', 'f', 'U', 'U', 'R', 'R',
'U', 'F', 'F', 'F', 'U', 'U', 'r', 'U', 'F', 'u', 'f', 'r', 'F', 'F', 'r', 'U', 'r', 'U', 'r',
'u', 'F', 'F', 'u', 'R', 'U', 'F', 'R', 'F', 'U', 'f', 'u', 'R', 'U', 'U', 'R', 'u', 'u', 'u',
'F', 'r', 'u', 'R', 'u', 'R', 'R', 'f', 'f', 'U', 'r', 'u', 'F', 'R', 'F', 'U', 'f', 'U', 'F',
'R', 'u', 'R', 'u', 'f', 'f']

Solving with BFS...

BFS solution: ['R', 'f', 'R', 'U', 'f', 'R', 'F', 'U', 'U', 'F']

Cost: 10

Nodes expanded: 303549

Time: 5.009856224060059

Solving with DFS (depth limit 15)...

DFS solution: ['F', 'F', 'F', 'R', 'F', 'R', 'f', 'R', 'U', 'f', 'R', 'F', 'U', 'F']

Cost: 14

Nodes expanded: 3698029

Time: 6.845629453659058

Solving with A*...

A* solution: ['R', 'f', 'R', 'U', 'f', 'R', 'F', 'U', 'U', 'F']

Cost: 10

Nodes expanded: 77636

Time: 2.741687774658203

```
In [7]: # Just for fun, let's also run DFS without a depth limit
dfs_no_limit = Solver(algorithms.dfs)
scramble_state, scramble_moves = dfs_no_limit.scramble(depth=15, rand_seed=42)
print("Scramble moves:", scramble_moves)
print()
print("Running DFS with no depth limit...")
start_time = time()
try:
    dfs_no_limit_solution, dfs_no_limit_nodes = dfs_no_limit.solve()
    dfs_no_limit_time = time() - start_time
    print("DFS no limit solution:", dfs_no_limit_solution)
    print("Cost:", len(dfs_no_limit_solution))
    print("Nodes expanded:", dfs_no_limit_nodes)
    print("Time:", dfs_no_limit_time)
except RecursionError:
    dfs_no_limit_time = time() - start_time
    print("Max recursion depth reached.")
    print("Time:", dfs_no_limit_time)
```

Current cube state: ROOGWGWYRWBBGBYOYGWORBYR

```

  R O
  O G
W G W Y R W B B
G B Y O Y G W O
  R B
  Y R
```

Scramble moves: ['u', 'F', 'F', 'u', 'R', 'R', 'R', 'u', 'F', 'u', 'u', 'r', 'F', 'r', 'f']

Running DFS with no depth limit...

Max recursion depth reached.

Time: 0.01898932456970215

In []: